

Testing Real Applications

Generating Test Cases Automatically

- Automation means using tools and algorithms to create test cases instead of writing them manually.
- During test case generation, the system generates test cases from requirements, models or code.

Generating Test Cases Automatically

Method	Description	Example	Test Case	Tools
Specification-based	Tool reads requirement and generates test scenarios and inputs.	"System should allow bursar to record student fees"	Valid amount Negative amount Zero Large amount	TestRail qTest
Model-Based	Uses diagrams that show workflow	E.g. Login → Validate → Success/Fail	Valid login Invalid login	TestComplete Spec Explorer

Generating Test Cases Automatically

Method	Description	Example	Test Case	Tools
Code-Based	Tool analyzes source code and generates test cases for branches, conditions, paths	If amount > 0 save() else displayError()	amount=50 amount = 100 amount = 0 amount = -20	JUnit pytest
Record and Playback	Tester performs actions once Tools records them and converts into test cases/ scripts	Actions Login Enter fees Submit	Tool generates reusable test script	Katalon Studio Selenium

Generating Test Cases Automatically

Method	Description	Example	Test Case	Tools
Using AI	AI reads UI or behavior and generates test steps automatically	Tool scans fees payment form	Button clicks Navigation	TestProject Katalon Studio Selenium

Difference between Test Case and Test Script

Test Case	A test case is a set of steps used to verify a feature works correctly. Includes inputs, steps, expected results used to test system.	Example: Enter student fee = 50,000 Click submit Expected: balance updated
Test Script	This is a set of instructions written in a programming language to automate testing. Steps that are automatically execute a test case.	Example: Open page Enter 50,000 Click button Verify updated balance

Converting Test Case into Test Script

- Test cases can be converted into pseudocode test scripts before writing code in a programming language.

Test Case Example

Test Case	Web FMS Login
Test Case ID	TC_LOGIN_01
Test Scenario	Verify valid user login
Input	Username: bursar Password: 1234
Test Steps	1. Open FMS 2. Enter username 3. Enter password 4. Click Login
Expected Result	User is redirected to dashboard

Example 1: Pseudocode for Test Script

BEGIN TEST Login

OPEN browser

NAVIGATE TO "http://fms-school.com/login"

ENTER "bursar" INTO username field

ENTER "1234" INTO password field

CLICK login button

VERIFY page contains "Dashboard"

CLOSE browser

END TEST

Step Mapping

Action	Meaning
OPEN browser	Start test environment
NAVIGATE	Load app
ENTER	Input data
CLICK	Perform action
VERIFY	Check expected result
CLOSE	End test

Reusable Script

BEGIN TEST LOGIN

ENTER username

ENTER password

CLICK login

IF dashboard displays

MARK test as **PASS**

ELSE

MARK test as **FAIL**

END TEST

Using Katalon Studio

Open browser

Navigate To URL "http://fms-school.com/login"

Set Text username_field bursar

Set Encrypted Text password_field 1234

Click login_button

Verify Element Present dashboard_Element

Close browser

Example 2: Pseudocode

- Accepts valid payment
- Rejects invalid payment
- Updates balance

BEGIN TEST FeesPayment

OPEN browser

NAVIGATE TO "<http://fms-school.com/feesPaymentForm>"

INPUT studentID

GET balance FROM Database

INPUT amount

IF amount > 0 AND amount <= balance **THEN**

 balance = balance – amount

DISPLAY "Payment Successful"

ELSE

DISPLAY "Invalid Payment"

END IF

CLOSE browser

END TEST



END